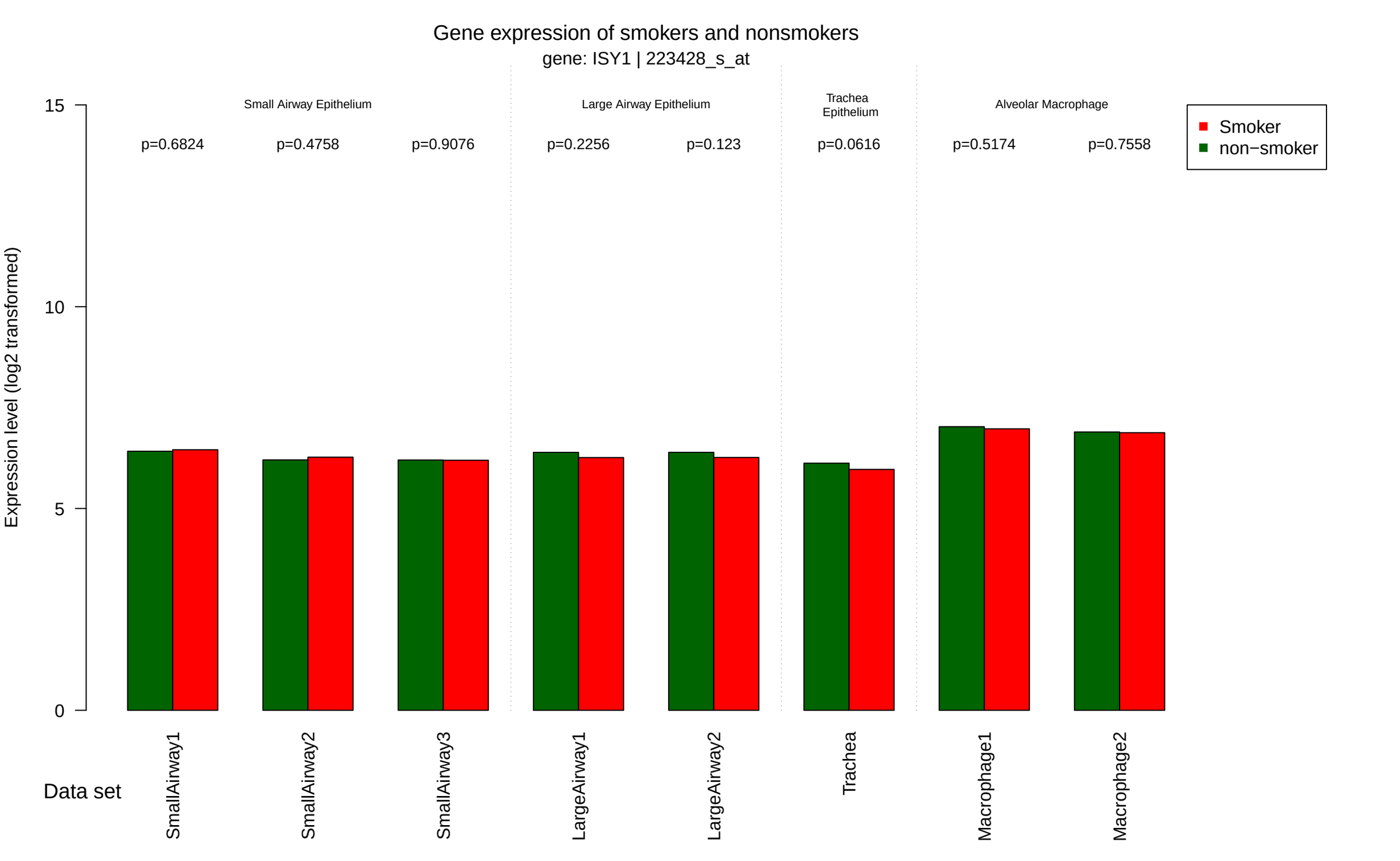
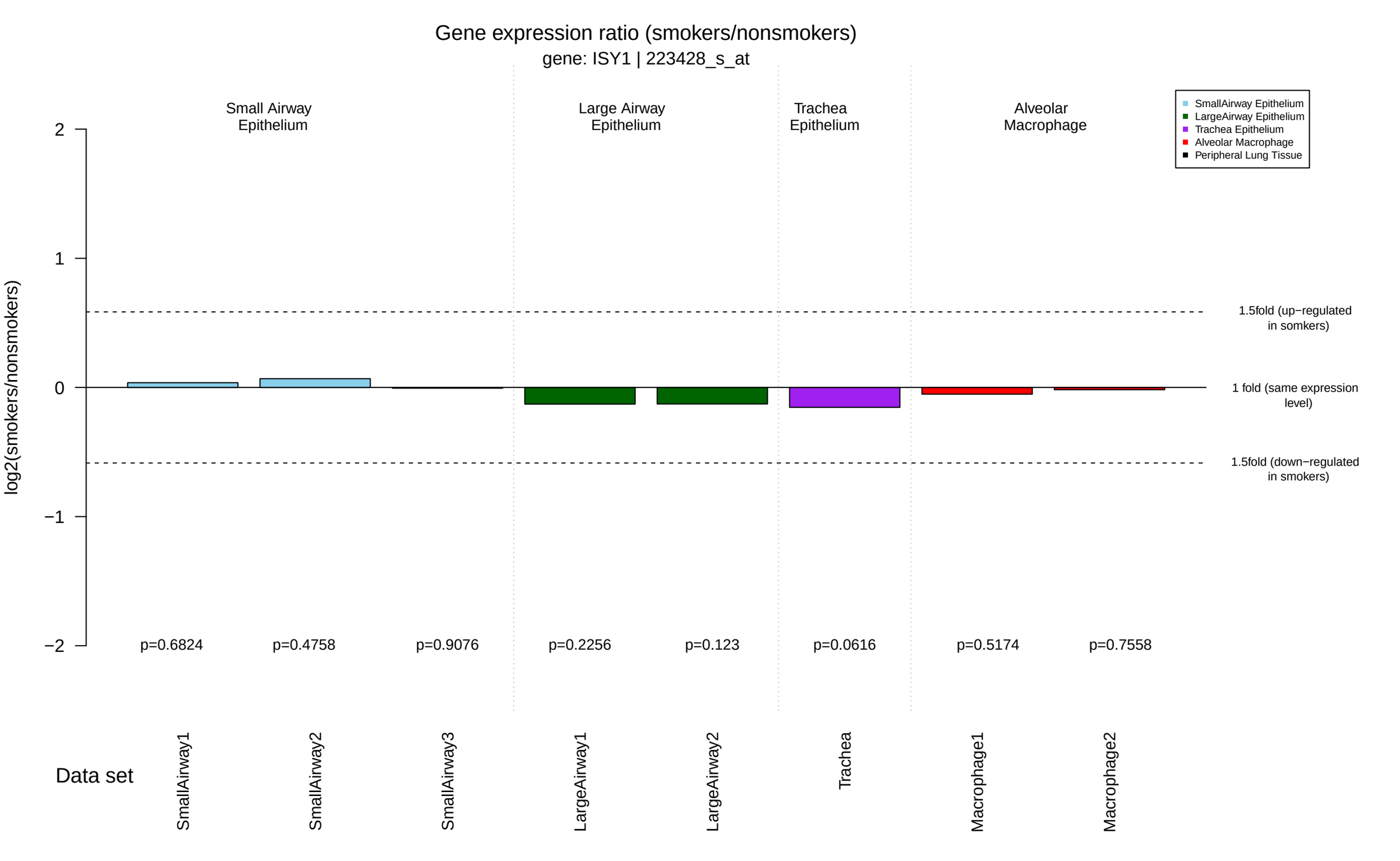
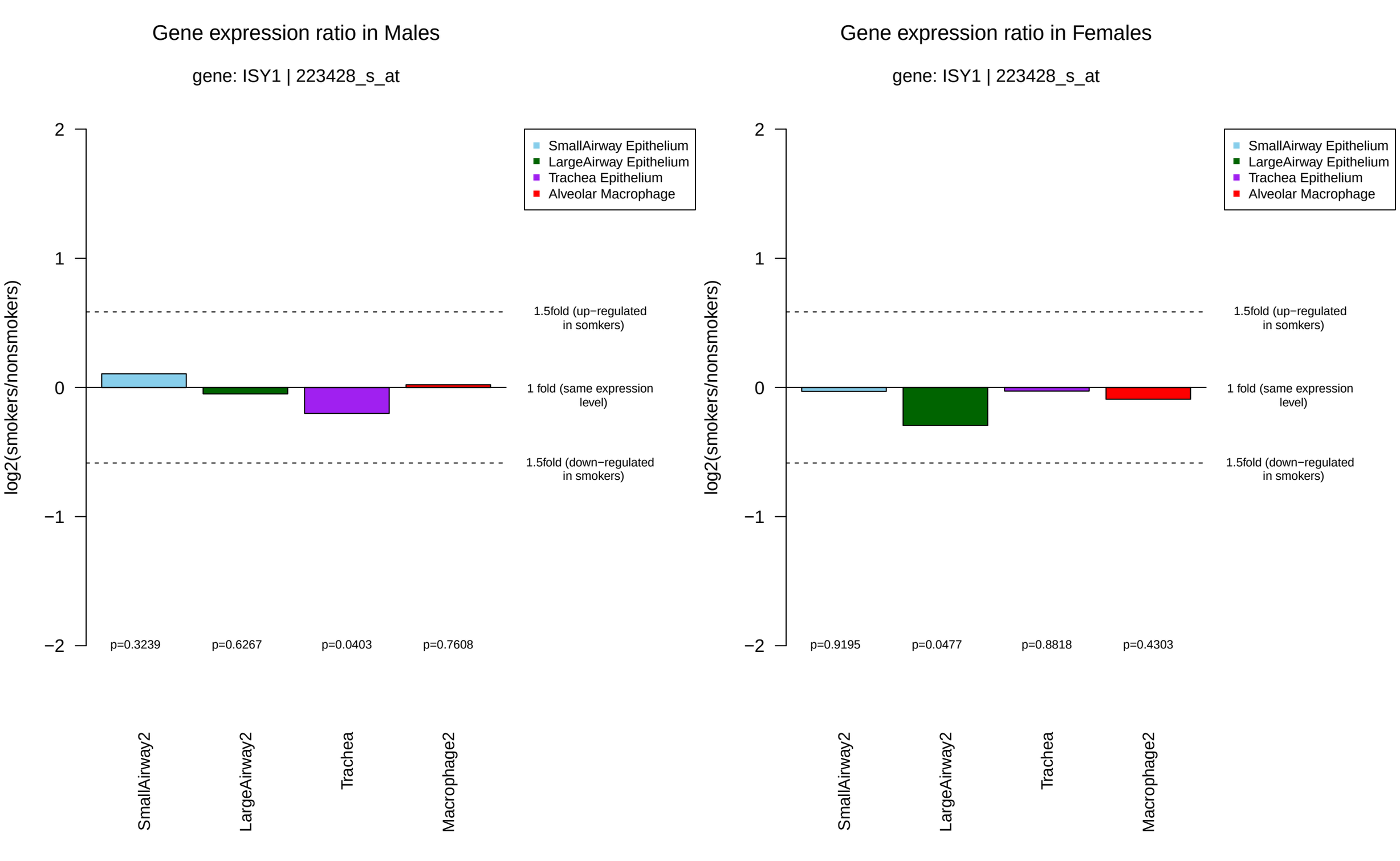
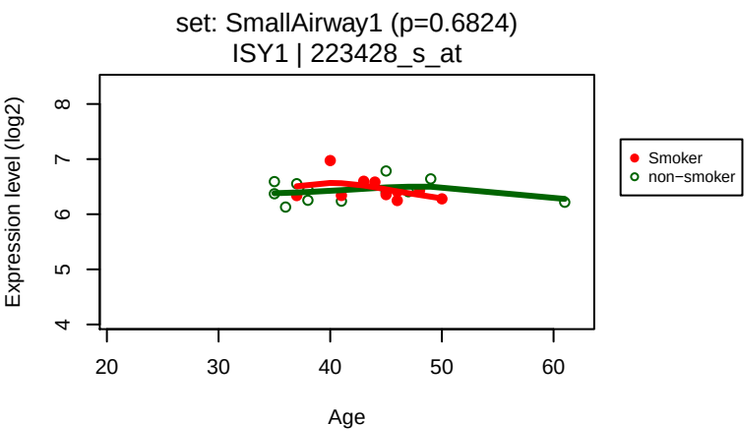
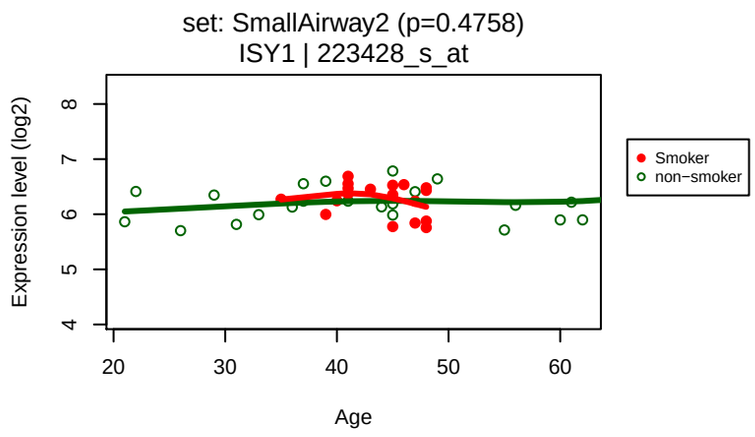
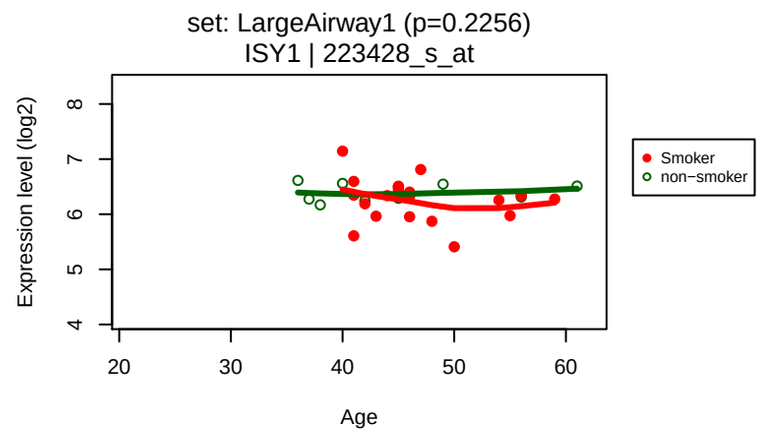

Gene expression of smokers and nonsmokers
gene: ISY1 | 223428_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6824
p=0.4758
p=0.9076
p=0.2256
p=0.123
p=0.0616
p=0.5174
p=0.7558
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

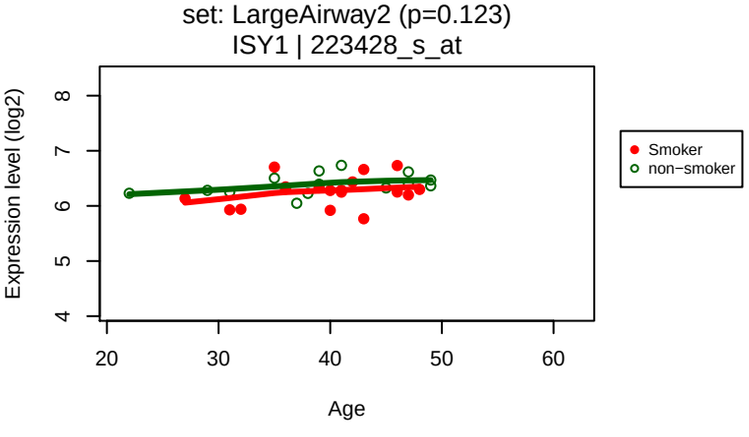

Gene expression ratio (smokers/nonsmokers)
gene: ISY1 | 223428_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6824
p=0.4758
p=0.9076
p=0.2256
p=0.123
p=0.0616
p=0.5174
p=0.7558
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

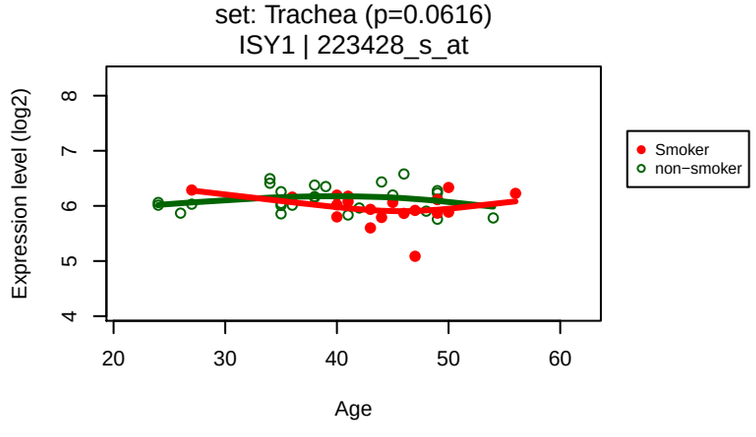

Gene expression ratio in Males
gene: ISY1 | 223428_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3239
p=0.6267
p=0.0403
p=0.7608
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ISY1 | 223428_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9195
p=0.0477
p=0.8818
p=0.4303
SmallAirway2
LargeAirway2
Trachea
Macrophage2

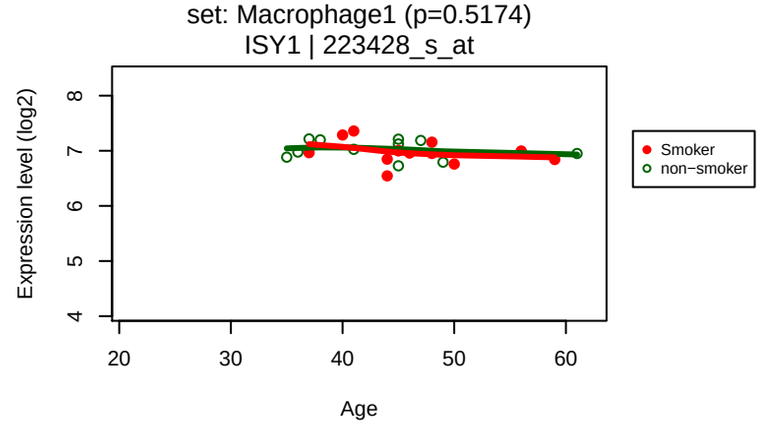

set: SmallAirway1 (p=0.6824)
ISY1 | 223428_s_at
Expression level (log2)
Age
Smoker
non−smoker

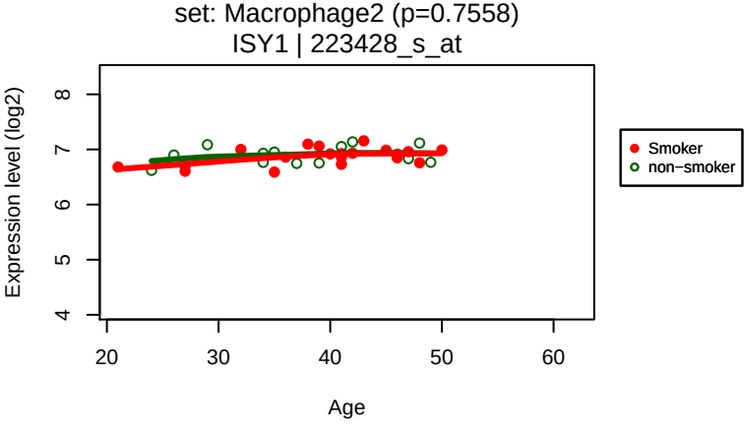

set: SmallAirway2 (p=0.4758)
ISY1 | 223428_s_at
Expression level (log2)
Age
Smoker
non−smoker

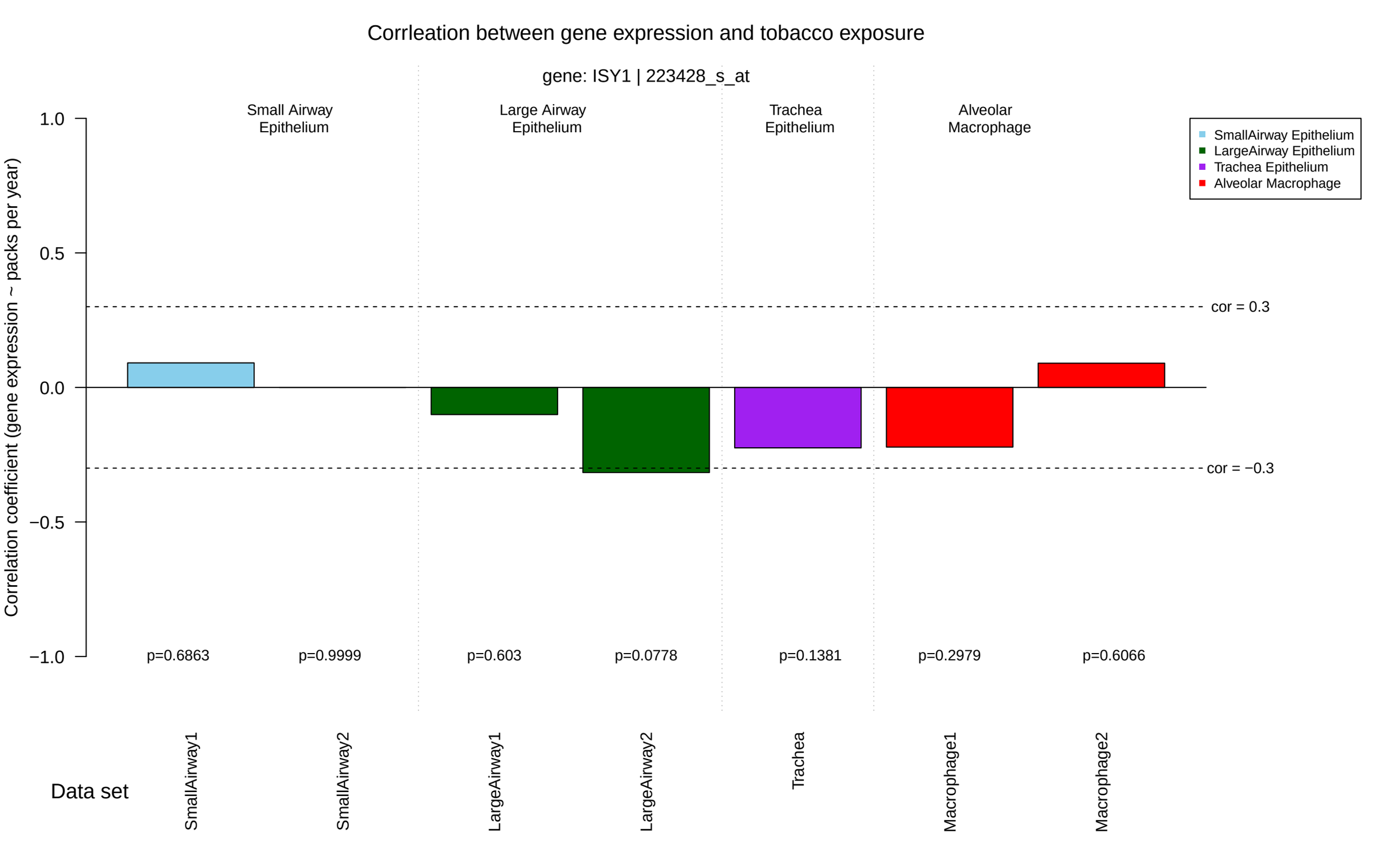

set: LargeAirway1 (p=0.2256)
ISY1 | 223428_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.123)
ISY1 | 223428_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0616)
ISY1 | 223428_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5174)
ISY1 | 223428_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7558)
ISY1 | 223428_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ISY1 | 223428_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6863
p=0.9999
p=0.603
p=0.0778
p=0.1381
p=0.2979
p=0.6066
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2